

AI-Based Medical Diagram Assessment System – Lecturer Questionnaire

This questionnaire is part of a final year project aimed at developing an AI-based system for automatically assessing **medical diagrams** (e.g., anatomy, histology, pathology diagrams).

The goal is to improve grading efficiency, consistency, and quality of feedback in medical education.

Your responses will help define system requirements, grading criteria, and design decisions.

All responses will remain confidential and will be used strictly for academic purposes.

* Indicates required question

Lecturer Profile

1. What is your teaching role? *

Mark only one oval.

- ☐ Lecturer
- ☐ Tutorial Instructor
- ☐ Clinical Instructor
- ☐ Teaching assistant
- ☐ Other:

2. What is your area of specialization? *

Mark only one oval.

- ☐ Anatomy
- ☐ Physiology
- ☐ Pathology
- ☐ Histology
- ☐ Surgery
- ☐ Pharmacology
- ☐ Other:

3. How many years of teaching experience do you have? *

Mark only one oval.

- ☐ Less than 2 years
- ☐ 2-5 years
- ☐ 6-10 years
- ☐ more than 10 years

Use of Medical Diagrams in Assessment

4. How often do you include medical diagram questions in assessments? *

Mark only one oval.

- ☐ Very frequently
- ☐ Often
- ☐ Sometimes
- ☐ Rarely
- ☐ Never

5. What types of medical diagrams do you assess? (Select all that apply) *

Check all that apply.

- ☐ Anatomy diagrams
- ☐ Histology diagrams
- ☐ Physiological/ process diagrams
- ☐ Pathology diagrams
- ☐ Clinical / surgical diagrams
- ☐ Other: _____

6. What skills you expect students to demonstrate in diagram-based questions?(Select *
all that apply)

Check all that apply.

- ☐ Accurate drawing of structures
- ☐ Correct labeling
- ☐ Understanding relationships between structures
- ☐ Completeness of the diagram
- ☐ Clarity and presentations

Current Grading Practices

7. How do you currently grade medical diagrams? *

Mark only one oval.

- ☐ Manually using a rubric
- ☐ Manually without a formal rubric
- ☐ Assisted by teaching assistants
- ☐ Other:

8. Do you use a standard marking rubric for diagrams? *

Mark only one oval.

- ☐ Yes
- ☐ No
- ☐ Sometimes

9. How do you allocate marks for diagram-based questions? *

Mark only one oval.

- ☐ Fixed marks per labelled parts
- ☐ Overall impression-based scoring
- ☐ Combination of both
- ☐ Other:

10. Do you provide partial marks for partially correct diagrams? *

Mark only one oval.

- ☐ Always
- ☐ Often
- ☐ Sometimes
- ☐ Rarely
- ☐ Never

11. How do you handle incorrect but closely related answers (e.g., correct region but wrong label)? *

Mark only one oval.

- ☐ Award partial marks
- ☐ Deduct full marks
- ☐ Depend on context
- ☐ Other:

12. On average, how long does it take to grade one diagram-based answer? *

Mark only one oval.

- ☐ less than 1 minute
- ☐ 1-3 minutes
- ☐ 3-5 minutes
- ☐ more than 5 minutes

13. How consistent do you think diagram grading is across different graders? *

Mark only one oval.

- ☐ Very consistent
- ☐ Moderately consistent
- ☐ Slightly consistent
- ☐ Not consistent

14. How often do you re-evaluate or adjust marks after initial grading *

Mark only one oval.

- ☐ Frequently
- ☐ Occasionally
- ☐ Rarely
- ☐ Never

Challenges in Grading Medical Diagrams

15. **What challenges do you face when grading medical diagrams? (Select all that apply)** *

Check all that apply.

- ☐ Time consuming process
- ☐ Difficulty interpreting drawings
- ☐ Inconsistent labeling
- ☐ Subjectivity in marking
- ☐ Large number of scripts
- ☐ Other: _____

16. **Which of the following contributes most to grading inconsistency? (Select all that apply)**

Check all that apply.

- ☐ Human subjectivity
- ☐ Fatigue during marking
- ☐ Lack of standardized rubric
- ☐ Variation in drawing quality
- ☐ Time pressure

17. How difficult is it to ensure fairness in grading diagram-based answers

Mark only one oval.

	1	2	3	4	5	
Very	☐	☐	☐	☐	☐	Very difficult

18. Have you experienced inconsistencies in grading between different examiners

Mark only one oval.

- ☐ Yes
- ☐ No
- ☐ Not sure

Feedback Practices

19. Do you usually provide feedback on medical diagram answers? *

Mark only one oval.

- ☐ Yes
- ☐ Sometimes
- ☐ Rarely
- ☐ Never

20. What type of feedback do you typically provide(*Select all that apply*) *

Check all that apply.

- ☐ Written comments
- ☐ Marks only
- ☐ Correct diagram reference
- ☐ Verbal feedback
- ☐ None

21. How effective do you think your feedback is in helping students improve? *

Mark only one oval.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ Very effective

22. How often are you able to give individualized feedback? **Mark only one oval.*

- ☐ Always
- ☐ Often
- ☐ Sometimes
- ☐ Rarely
- ☐ Never

23. What challenges do you face in providing feedback? (Select all that apply) **Check all that apply.*

- ☐ Lack of time
- ☐ Large class sizes
- ☐ Difficulty explaining errors
- ☐ Lack of tools
- ☐ Other: _____

24. Would automated feedback tools improve your teaching workflow? **Mark only one oval.*

	1	2	3	4	5	
Not	☐	☐	☐	☐	☐	Very helpful

AI-Based Model Expectations

25. **Do you think an automated model could assist in grading medical diagrams?** *

(Scale: 1–5)

Mark only one oval.

1 2 3 4 5

Strongly disagree ☐ ☐ ☐ ☐ ☐ Strongly agree

26. **In which areas should AI assist most? (Select all that apply) ***

Check all that apply.

- ☐ Identify anatomical structures
- ☐ Checking labels
- ☐ Scoring diagrams
- ☐ Providing feedback
- ☐ Detecting missing components

27. **How important is explainability (clear reasoning behind scores)? ***

(Scale: 1–5)

Mark only one oval.

1 2 3 4 5

Not important ☐ ☐ ☐ ☐ ☐ Very important

28. Would you trust an AI system to assist in grading? *

Mark only one oval.

- ☐ Yes
- ☐ No
- ☐ Maybe

29. **What concerns do you have about automated grading?** (Select all that apply) *

Check all that apply.

- ☐ Incorrect grading
- ☐ Lack of clinical judgement
- ☐ Misinterpretation of diagrams
- ☐ Reliability issues
- ☐ Data privacy

30. What level of control would you prefer? *

Mark only one oval.

- ☐ Full manual control
- ☐ Partial(AI + human validation)
- ☐ Fully automated
- ☐ Not sure

31. How important is manual override of AI scores?

Mark only one oval.

	1	2	3	4	5	
Not	☐	☐	☐	☐	☐	Very important

Medical Accuracy & Trust

32. How critical is accuracy in identifying anatomical structures? *

Mark only one oval.

	1	2	3	4	5	
Not	☐	☐	☐	☐	☐	Extremely critical

33. What is the acceptable margin of error? *

Mark only one oval.

- ☐ Very low(almost none)
- ☐ Low
- ☐ Moderate
- ☐ High

34. **What would increase your trust in the system? (Select all that apply) ***

Check all that apply.

- ☐ High accuracy
- ☐ Transparent grading
- ☐ Ability to review decisions
- ☐ Consistency
- ☐ Validation against human grading

35. **Would you be willing to compare AI grading with your own for validation? ***

Mark only one oval.

- ☐ Yes
- ☐ No
- ☐ Maybe

System Requirements Input

36. **What criteria should be used when grading medical diagrams? ***

37. How should marks be distributed across diagram components? *

38. What output format would you prefer? (Select all that apply) *

Check all that apply.

- ☐ Numerical score
- ☐ Breakdown of marks
- ☐ Annotated diagram
- ☐ Written feedback
- ☐ Summary report

39. Should the system integrate with Learning Management Systems (LMS)? *

Mark only one oval.

- ☐ Yes
- ☐ No
- ☐ Not ne

40. What level of accuracy do you expect from the system?

Mark only one oval.

- ☐ Below 70%
- ☐ 70-80%
- ☐ 80-90%
- ☐ above 90%

Final feedback

41. What improvements would you suggest for an automated diagram grading system?

42. In your opinion, what would make this system truly useful in teaching?

43. **Any additional comments or recommendations?** *(Optional)*

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